MSQF 533: FINANCIAL RISK MANAGEMENT

UNIT II – VaR METHODS AND HEDGING

1. Introduction to Value at Risk (VaR)

1.1 Meaning of financial risk

Financial risk is the possibility of losing money because prices, interest rates, exchange rates or other financial conditions change. For example, if a person buys shares worth ₹1,00,000 and their value falls by 10%, the loss is ₹10,000. This is market risk.

1.2 What is Value at Risk?

Value at Risk (VaR) is a statistical measure that estimates a loss threshold over a specified period at a chosen confidence level, under the assumptions of a model. In simple words, it estimates how much a portfolio could lose in ordinary market conditions over a given period.

Example: A bank reports a one-day VaR of ₹5 lakh at 95% confidence. This means the model estimates that the loss should not exceed ₹5 lakh on about 95% of comparable trading days. On about 5% of days, the loss may be greater. VaR is not a guarantee and is not the maximum possible loss.

1.3 Confidence level and tail probability

Confidence level	Tail probability	Simple interpretation
95%	5%	Model expects the loss threshold to be exceeded on about 5% of comparable days.
99%	1%	Model expects the loss threshold to be exceeded on about 1% of comparable days.

A higher confidence level usually produces a higher VaR because the measure focuses on a more extreme loss threshold.

1.4 Main components of VaR

Component	Meaning	Example
Portfolio value	Current value of investments	₹10,00,000
Time horizon	Period being measured	One day
Confidence level	Chosen probability level	95%
VaR amount	Estimated loss threshold	₹20,000

1.5 Uses and limitations of VaR

- Measures and monitors market risk.
- Helps set trading limits and compare portfolios.
- Supports risk reporting and capital planning.
- Depends on data, assumptions and the chosen time horizon.
- Does not explain how large losses may be after the VaR threshold is crossed.
- May underestimate risk during crises or when markets behave differently from the past.

2. Downside VaR

2.1 Meaning

In everyday conversation, "downside risk" means the risk of losing money rather than the possibility of making a profit. Downside-focused risk measures pay special attention to negative returns. In formal risk literature, the term "VaR" already refers to a loss quantile; "Downside VaR" is sometimes used in course materials to emphasize losses below a chosen target or to distinguish downside-focused analysis from measures that consider variation on both sides of the average. The exact convention should be stated when using the term.

2.2 Simple example

An investment has daily returns of +2%, +1%, −1%, −2% and −5%. A downside-focused analysis concentrates on the negative returns (−1%, −2%, −5%) because they represent losses. It asks how severe the losses may become, rather than treating unusually high gains as equally risky.

2.3 Why downside risk matters

- Investors usually worry more about losses than unexpected gains.
- A retirement fund may need to protect capital.
- A bank must estimate potential losses on its trading portfolio.
- A company may need to plan for a fall in foreign-currency receipts.

3. Main VaR Methods

Different methods estimate VaR in different ways. Some use simplified mathematical assumptions; others revalue the portfolio using historical or simulated market changes.

Method	Basic idea	Useful when	Main limitation
Delta-normal / parametric	Uses volatility, correlations and an assumed return distribution.	Risk factors are relatively well behaved and positions are approximately linear.	Assumptions may fail during extreme markets.
Historical simulation	Applies actual past market movements to today's portfolio.	Historical data are available and relevant.	Past events may not represent future conditions.
Monte Carlo simulation	Creates many possible market scenarios using a statistical model.	Complex portfolios and many risk factors need analysis.	Results depend on model design and computing assumptions.

3.1 Local valuation and full valuation

Local valuation

Local valuation estimates the effect of small changes in market factors using a simplified approximation. It looks near the current market position rather than recalculating every instrument in full for every scenario. Delta-based approaches are common examples.

Example: If a share portfolio is worth ₹10 lakh and a small market move is expected to reduce its value by approximately 2%, a local approximation estimates a loss of ₹20,000.

Full valuation

Full valuation recalculates the value of each instrument under each market scenario. It is more detailed and can capture non-linear price behaviour, but it may require more data and computing time.

Example: For an options portfolio, the risk team changes the assumed share price and volatility, then reprices each option using an appropriate pricing model. This can capture effects that a simple percentage approximation may miss.

Point	Local valuation	Full valuation
Calculation	Approximation around current market conditions	Reprices instruments under each scenario
Speed	Usually faster	Usually more computationally intensive

Non-linear products	May be less accurate unless extended	Generally better at capturing non-linear behaviour
Example	Delta approximation	Repricing options under many share-price scenarios

4. Delta-Normal Method

4.1 Meaning

The delta-normal method, also called a variance-covariance approach in common applications, estimates VaR using the sensitivity of a portfolio to market changes, the volatility of those changes, and an assumption that returns are approximately normally distributed.

"Delta" means how much an instrument's value changes when an underlying market factor changes. "Normal" refers to the bell-shaped probability distribution assumed by the method.

4.2 Basic formula for a simple portfolio

For a simple portfolio with normally distributed returns, a commonly used approximation is:

VaR ≈ Portfolio value × volatility × z-value

Here, volatility is measured over the chosen time horizon and the z-value corresponds to the selected confidence level. Common one-tailed z-values are about 1.645 for 95% confidence and 2.326 for 99% confidence.

4.3 Worked example

A portfolio is worth ₹10,00,000. Its estimated one-day standard deviation of return is 1.2%. Estimate one-day 95% VaR using z = 1.645.

VaR ≈ ₹10,00,000 × 0.012 × 1.645 = ₹19,740 (approximately).

Interpretation: Under the model assumptions, the estimated one-day 95% loss threshold is about ₹19,740.

4.4 Advantages and limitations

Advantages

- Relatively quick and easy to calculate.
- Uses volatility and correlations to summarize market risk.
- Convenient for portfolios with approximately linear exposures.

Limitations

- The normal-distribution assumption may underestimate extreme events.

- Non-linear products such as options can be difficult to represent accurately using delta alone.
- Results depend on the quality of volatility and correlation estimates.

5. Historical Simulation

5.1 Meaning

Historical simulation estimates VaR by looking at actual market movements observed in a past period and applying those movements to the current portfolio. It does not need to assume that returns follow a normal distribution.

5.2 Steps

1. Collect historical changes in share prices, interest rates, exchange rates or other risk factors.
2. Apply each historical change to today's portfolio.
3. Calculate the profit or loss under each historical scenario.
4. Arrange the simulated losses from smallest to largest.
5. Select the loss at the chosen tail percentile to estimate VaR.

5.3 Simple numerical illustration

Suppose a simplified exercise uses 100 historical daily portfolio losses. After sorting the losses from smallest to largest, the loss near the 95th percentile is ₹25,000. The historical-simulation estimate of 95% one-day VaR is approximately ₹25,000, subject to the percentile convention used.

This means roughly 5% of the observations lie beyond that loss threshold. With 100 observations, the estimate is sensitive to the chosen ranking convention and the sample period.

5.4 Advantages and limitations

- Advantage: It uses real historical market patterns and can reflect non-normal outcomes present in the sample.
- Advantage: It is relatively easy to explain.
- Limitation: A historical period may omit a crisis or may include events that are no longer representative.
- Limitation: The method can be slow for large, complex portfolios if each scenario requires full repricing.

6. Monte Carlo Simulation

6.1 Meaning

Monte Carlo simulation estimates risk by generating a large number of possible future market scenarios using a statistical model. Each scenario is used to calculate a possible portfolio profit or loss.

Imagine asking a computer to create thousands of plausible versions of tomorrow's market. The computer reprices the portfolio in each version and builds a distribution of possible outcomes.

6.2 Steps

6. Choose the risk factors to model, such as share prices, interest rates or exchange rates.
7. Estimate the statistical behaviour of those factors and how they move together.
8. Generate many random market scenarios.
9. Revalue the portfolio in every scenario.
10. Sort the resulting losses and select the required percentile.

6.3 Example

A bank models 10,000 possible one-day market scenarios. The portfolio losses are sorted from lowest to highest. If the loss at the 95th percentile is ₹30,000, the estimated 95% one-day VaR is approximately ₹30,000.

6.4 Advantages and limitations

- Can model many risk factors and complex instruments.
- Can be adapted to different distributions and relationships.
- Requires assumptions about the market and the relationships among risk factors.
- Can require significant computing power and model validation.
- A very large number of simulations does not fix a badly designed model.

7. Comparing the Three Major VaR Approaches

Feature	Delta-normal	Historical simulation	Monte Carlo simulation
Data basis	Volatility, correlations and distribution assumptions	Observed historical changes	Model-based generated scenarios
Distribution assumption	Often assumes normality	Uses the historical sample distribution	Chosen by the modeller
Complexity	Low to moderate	Moderate	Moderate to high
Main strength	Fast and relatively simple	Intuitive and based on observed data	Flexible for complex portfolios
Main weakness	Sensitive to assumptions	Depends on past data	Sensitive to model assumptions

8. Applications of VaR

VaR is a decision-support tool, not a guarantee. Common applications include:

- Trading limits: A bank may limit the VaR that a trading desk can take.

- Portfolio comparison: Managers can compare the estimated market risk of different portfolios.
- Risk reporting: Senior management can receive a consistent summary of market risk.
- Capital planning: VaR-related measures may support internal risk and capital processes, subject to applicable regulation.
- Performance evaluation: Managers can consider returns alongside the amount of risk taken.
- Monitoring: A sudden increase in VaR can signal that positions have become more volatile or concentrated.

Example

A securities firm sets a daily VaR limit of ₹50 lakh for a trading desk. If its calculated VaR rises to ₹65 lakh, the desk may need to reduce positions, hedge exposures, or obtain approval under the firm's risk policy. The limit is an internal control, not a prediction that losses cannot exceed ₹65 lakh.

9. Stress Testing

9.1 Meaning

Stress testing asks: "What might happen to our portfolio if a severe but plausible event occurs?" Unlike ordinary VaR, which focuses on a selected percentile under model assumptions, stress testing examines specific adverse scenarios that may be rare but damaging.

9.2 Types of stress testing

Type	Meaning	Example
Historical scenario	Recreates a past market crisis.	Apply market movements similar to a past stock-market crash.
Hypothetical scenario	Creates a severe event that has not necessarily occurred in that exact form.	Assume interest rates rise sharply and the currency weakens together.
Sensitivity analysis	Changes one risk factor to see its effect.	Check portfolio loss if interest rates rise by 1 percentage point.
Reverse stress test	Starts with a damaging outcome and asks what could cause it.	Identify scenarios that could make a bank breach its capital limit.

9.3 Worked example

A company has investments worth ₹5 crore. Its risk team tests a hypothetical 20% fall in the relevant market. A simplified estimate gives a loss of ₹1 crore if the portfolio moves exactly with that market. In practice, the team should revalue actual holdings because different investments may respond differently.

9.4 VaR versus stress testing

VaR	Stress testing
Estimates a loss threshold at a probability level.	Examines the effect of specified severe scenarios.
Often summarizes ordinary market risk statistically.	Focuses on unusual, extreme or system-wide conditions.
Can hide the size of losses beyond the threshold.	Shows the possible impact of selected shocks.
Useful for routine risk monitoring.	Useful for resilience planning and contingency decisions.

10. Hedging: Meaning and Purpose

10.1 What is hedging?

Hedging means taking a position that reduces the risk of an existing exposure. It is similar to buying insurance: the aim is to reduce the effect of an unfavourable change, not necessarily to earn an additional profit.

10.2 Everyday example

An importer must pay US$10,000 in three months. If the US dollar becomes more expensive in Indian rupees, the importer's cost will rise. The importer can use a suitable forward contract to agree on an exchange rate now, reducing uncertainty about the future payment.

10.3 Common hedging instruments

Instrument	Simple explanation	Example use
Forward contract	Private agreement to buy or sell an asset at a future date for an agreed price.	Importer fixes a future exchange rate.
Futures contract	Standardized exchange-traded contract.	A wheat producer hedges the future selling price.
Options	Give the buyer a right, but not an obligation, to buy or sell at a specified price.	Investor buys a put option to protect shares from a fall.
Swaps	Agreement to exchange cash flows under specified terms.	Company swaps floating-rate payments for fixed-rate payments.

10.4 Benefits and costs of hedging

- Reduces uncertainty about future cash flows or portfolio value.
- Can protect profit margins and support budgeting.
- May involve premiums, transaction costs, margin requirements or imperfect matching.
- Can reduce gains if the market moves favourably.
- Does not always remove all risk; some residual risk may remain.

11. Linear Risk and Non-linear Risk

11.1 Linear risk

A position has approximately linear risk when its value changes in a fairly proportional way as the underlying market price changes. For example, ignoring fees and tracking differences, owning 100 shares means a ₹1 fall per share reduces the holding's value by about ₹100.

11.2 Non-linear risk

A position has non-linear risk when its value does not change in a constant proportion to the underlying market factor. Options are a common example: an option's sensitivity can change as the share price, time to expiry and volatility change.

Feature	Linear exposure	Non-linear exposure
Response to market move	Approximately proportional	May curve or change in sensitivity
Example	Holding ordinary shares	Holding a call or put option
Risk measurement	Sensitivity or delta may be adequate for small moves	May require full valuation and additional sensitivities
Hedging challenge	Hedge amount can remain relatively stable for small moves	Hedge may need frequent adjustment

12. Optimal Hedging

12.1 Meaning

Optimal hedging means choosing a hedge size that reduces risk effectively while considering cost, liquidity, and the type of exposure. A hedge that is too small leaves substantial risk; a hedge that is too large can create a new risk in the opposite direction.

12.2 Basic hedge ratio

A simple hedge ratio expresses how much of the exposure should be hedged. For a straightforward matched exposure, a 1:1 hedge may be suitable. When the hedge instrument does not move exactly with the exposure, a different ratio may be more effective.

12.3 Example

A company expects to sell 100 tonnes of a commodity in three months. It uses futures to hedge the selling price. If one futures contract covers 10 tonnes and the company wants to hedge the full quantity, the simple matched hedge is 10 contracts. Real hedges may differ because of contract specifications, timing, grade, location and basis risk.

12.4 Factors affecting the optimal hedge

- Size and timing of the exposure.
- How closely the hedge instrument tracks the exposure.
- Volatility of the exposure and hedge instrument.
- Transaction costs and liquidity.
- The organisation's tolerance for risk.
- Contract size and expiry date.

13. Hedge Ratio as a Regression Coefficient

13.1 Meaning

A regression-based hedge ratio estimates how much the hedge instrument tends to move when the asset being hedged moves. It is often estimated by regressing changes in the spot price on changes in the futures price, with the exact regression setup depending on the convention used.

For a common minimum-variance hedge, the hedge ratio is:

$h^* = \mathrm{Cov}(\Delta S, \Delta F) / \mathrm{Var}(\Delta F)$

Where ΔS is the change in the spot price, ΔF is the change in the futures price, Cov is covariance, and Var is variance. This formula estimates the futures position that minimizes the variance of the hedged price change under the model.

13.2 Numerical example

Suppose the covariance between spot-price changes and futures-price changes is 18, and the variance of futures-price changes is 24 (in consistent units).

$h^* = 18 / 24 = 0.75$.

Interpretation: The estimated hedge ratio is 0.75. Under the assumptions and units of this example, the hedge size is about 75% of the simple one-for-one exposure measure. The actual number of contracts also depends on the exposure size and futures contract size.

13.3 Why use regression?

- The spot and futures prices may not move exactly together.
- Historical data can help estimate their relationship.
- The hedge ratio can reduce variability more effectively than automatically choosing a 1:1 hedge.
- The relationship may change, so estimates should be reviewed.

14. Hedging Non-linear Risk

14.1 Why is non-linear hedging different?

For non-linear instruments such as options, the amount of risk changes when market conditions change. A hedge that works today may be too large or too small tomorrow. Hedgers may need to adjust their positions as prices, volatility and time to expiry change.

14.2 Delta hedging

Delta measures how much an option's price is expected to change for a small change in the underlying asset price. Delta hedging uses the underlying asset or another instrument to offset this first-order sensitivity.

Example: If an option position has a total delta of +500 shares, a trader may sell approximately 500 shares to make the combined position delta-neutral at that moment. This is only a simplified illustration; contract multipliers and the option position's actual delta must be included in practice.

14.3 Gamma and rebalancing

Gamma measures how quickly delta changes as the underlying price changes. A portfolio with significant gamma may not remain delta-neutral after the market moves. The trader may need to rebalance the hedge.

14.4 Vega and volatility risk

Vega measures how sensitive an option's value is to changes in implied volatility. A portfolio may lose value even if the underlying price does not move much, simply because volatility changes.

14.5 Practical example

An investor owns call options on a company's shares. The investor sells shares to offset the options' current delta. If the share price moves sharply, the options' delta changes because of gamma. The investor may need to buy or sell more shares. If implied volatility changes, the option value can also change, creating vega risk that a share-only hedge does not fully remove.

14.6 Key approaches

- Delta hedging: offsets small price changes at the current point.
- Gamma management: helps manage changes in delta.
- Vega hedging: offsets some exposure to volatility changes.

- Full revaluation: reprices the option under different scenarios.
- Scenario analysis and stress testing: checks the hedge under large market moves.

15. Case Studies

Case Study 1: A bank uses VaR to control trading risk

Situation: A bank's equity trading portfolio is worth ₹20 crore. Its one-day 99% VaR is ₹40 lakh.

Interpretation: Under the model, the bank estimates that the loss should not exceed ₹40 lakh on about 99% of comparable trading days. The loss could exceed that amount on the remaining days.

Decision: The risk manager monitors the VaR limit, checks large positions and correlations, and asks for a review if the VaR increases sharply.

Lesson: VaR is useful for daily monitoring, but the bank should also use stress tests because a severe market crash can cause losses much larger than the VaR estimate.

Case Study 2: Historical simulation versus Monte Carlo

Situation: An investment fund owns shares, bonds and options. Historical simulation produces a 95% VaR of ₹18 lakh, while Monte Carlo simulation produces ₹23 lakh.

Interpretation: The difference does not automatically mean that one method is wrong. The methods may use different data, assumptions, scenarios and valuation techniques.

Decision: The fund examines the assumptions, validates the models, investigates the option exposures and compares the results with stress scenarios.

Lesson: Risk estimates should be understood and challenged, not accepted simply because they are numerical.

Case Study 3: An importer hedges foreign-exchange risk

Situation: An Indian importer must pay US$50,000 in 90 days. If the dollar strengthens, the rupee cost of the payment will rise.

Possible action: The importer enters an appropriate forward contract to lock in the exchange rate for the payment date.

Benefit: The importer can plan its rupee cash requirement more confidently.

Trade-off: If the dollar weakens, the importer may not benefit from the favourable exchange-rate movement in the same way it would without the hedge. Contract terms and counterparty risk also matter.

Lesson: Hedging reduces uncertainty, but it can have costs and may give up some favourable outcomes.

Case Study 4: A farmer hedges commodity-price risk

Situation: A farmer expects to harvest 50 tonnes of a crop in three months. The farmer is worried that market prices may fall before the crop is sold.

Possible action: If a suitable liquid futures contract exists and the farmer can access it, the farmer may sell futures to offset some of the risk of a falling cash-market price.

Limitation: The local crop price may not move exactly like the futures price. Differences in grade, location and timing create basis risk.

Lesson: A hedge should match the underlying exposure as closely as possible and should account for practical contract differences.

Case Study 5: An options trader manages non-linear risk

Situation: A trader holds call options. Initially, the trader sells shares to offset the options' positive delta.

Problem: The share price rises and the option delta increases. The original share hedge is no longer sufficient.

Action: The trader recalculates delta and adjusts the share position. The trader also monitors gamma and vega and runs stress tests.

Lesson: Non-linear exposures often require active monitoring and rebalancing; a one-time hedge may not be enough.

Term	Simple meaning
VaR	Estimated loss threshold at a chosen confidence level and time horizon.
Confidence level	Probability level used to define the VaR threshold.
Volatility	How much a price or return tends to vary.
Correlation	How two variables tend to move in relation to one another.
Delta	Approximate sensitivity of an instrument's value to a small change in an underlying factor.
Full valuation	Repricing the instrument under each scenario.
Stress testing	Checking the portfolio under severe specified conditions.
Hedging	Taking an offsetting position to reduce an existing risk.
Basis risk	Risk that the hedge and the exposure do not move exactly together.
Gamma	How quickly delta changes as the underlying price changes.
Vega	Sensitivity of an option's value to implied volatility.
Regression	A statistical method for estimating relationships between variables.